ASCORBIC ACID (VITAMIN C)

A description of scurvy was found in the Ebers papyrus written in 1500 B.C. in Egypt. During the voyage of Vasco da Gama, around the cape of Good Hope to India in 1498, he lost two-third of the crew due to scurvy. The French explorer, Jacques Cartier, in 1536, during the voyages to discover eastern parts of Canada, was laid up with scurvy. A friendly native gave an extract from the leaves of spruce tree, which produced remarkable cure to scurvy (Fig. 34.26). James Lind published 'Treatise on Scurvy', in 1753. These observations led to compulsory rationing of lime or lemon juice to all the crew of the British Roval Navv from 1795 onwards. So the

British sailors were nicknamed as "Limeys". However, it helped to eliminate scurvy from the British Navy, while opponents continued to suffer. No wonder, in course of time, Britain had the colonies in which the sun never set. In 1907, Holst and Trochlich produced scurvy in guineapigs. Zilva and associates, in 1928, showed that the antiscorbutic factor present in lemon juice is a reducing substance. The factor was isolated in 1930 and named as "Hexuronic acid" by Albert Szent-Gyorgi (Nobel prize, 1937). In 1933, Haworth established the molecular structure. He renamed it as ascorbic acid (Nobel prize, 1937).

Chemistry of Vitamin C

It is water soluble and is easily destroyed by heat, alkali and storage. In the process of cooking, 70% of vitamin C is lost.

The structural formula of ascorbic acid closely resembles that of carbohydrates (Fig. 34.28). The strong reducing property of vitamin C depends on the double-bonded (enediol) carbons.

Only L-ascorbic acid and dehydroascorbic acid have antiscorbutic activity. The D-ascorbic acid has no activity.

Biosynthesis of Ascorbic Acid in Animals

Most animals and plants can synthesize ascorbic acid from glucose. The pathway is described in Figure 10.7. **Man, higher primates**, guineapigs and bats are the only species which cannot synthesize ascorbic acid (block in gulono lactone oxidase step). They lack the genes responsible for the synthesis of this enzyme. The vitamin therefore should be supplied in the diet of these species. The staple diet of primates contains fruits and vegetables rich in ascorbic acid, and so the gene deletion will have no deleterious effect in primates. Human beings, of course, carried over this gene deletion.

Metabolism of Ascorbic Acid

i. Ascorbic acid is readily absorbed from gastrointestinal tract. The vitamin is excreted in urine. Since vitamin C is a strong reducing agent, the *Benedict's test* will be positive in the urine sample after the vitamin administration.

ii. Oxidation of ascorbic acid yields dehydro ascorbic acid, which is oxidized further to oxalic acid through diketo-L-gulonic acid (Fig. 34.28). Ascorbic acid is partly excreted unchanged and partly as oxalic acid. Most of the oxalates in urine are derived from ascorbic acid, and the rest from glycine metabolism.

iii. The ascorbic acid level varies between 0.7 to 1.2 mg/100 ml of plasma and 25 mg/100 cc of WBC. A low level in blood is noted in women taking contraceptive pills and also in chronic alcoholics.

iv. A very high concentration of vitamin C is observed locally in healing wounds. Vitamin C is essential for wound healing.

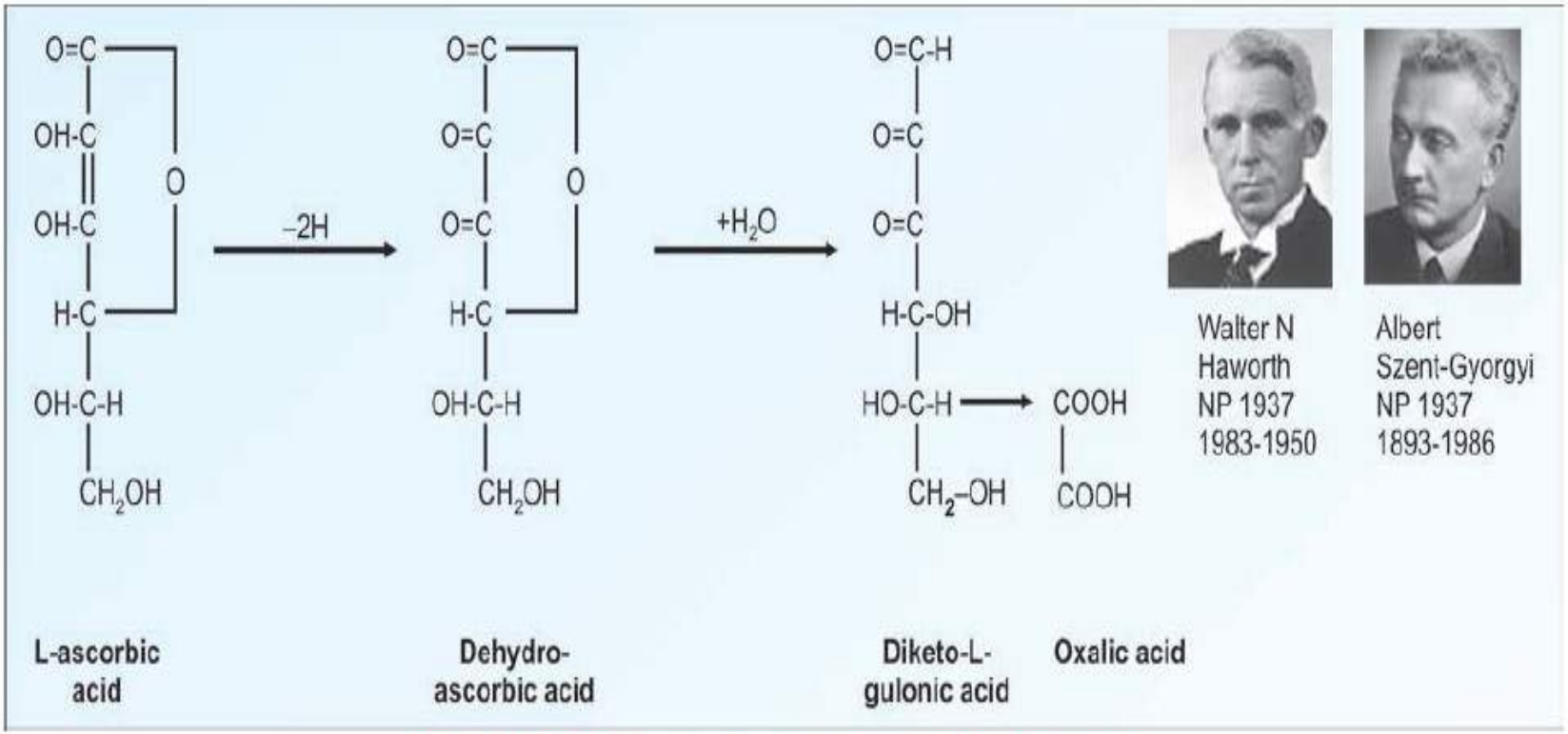


Fig. 34.28. Vitamin C: Structure and catabolism



Fig. 34.26. Red Indian gives the leaves to Cartier

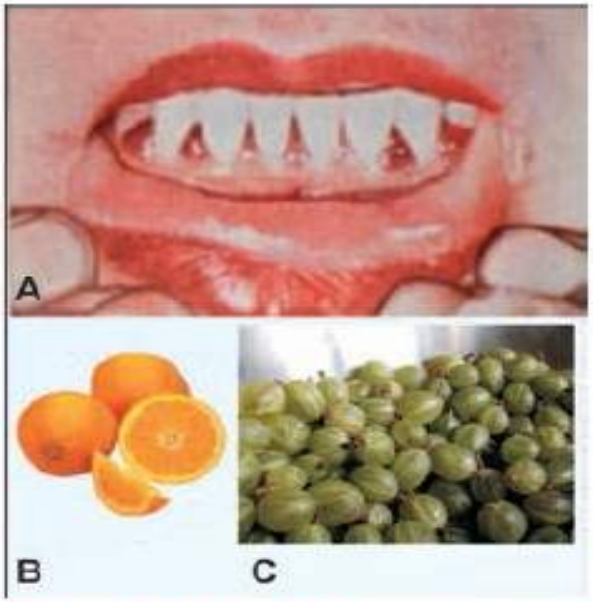


Fig. 34.27. (A) Gingivitis and bleeding gum in vitamin C deficiency. (B) Lime and (C) goosebery are good sources of vitamin C



Henry Dale
NP 1936
1875-1968



Otto Loewi
NP 1936
1873-1961



Bernard Katz
NP1970
1911-2003

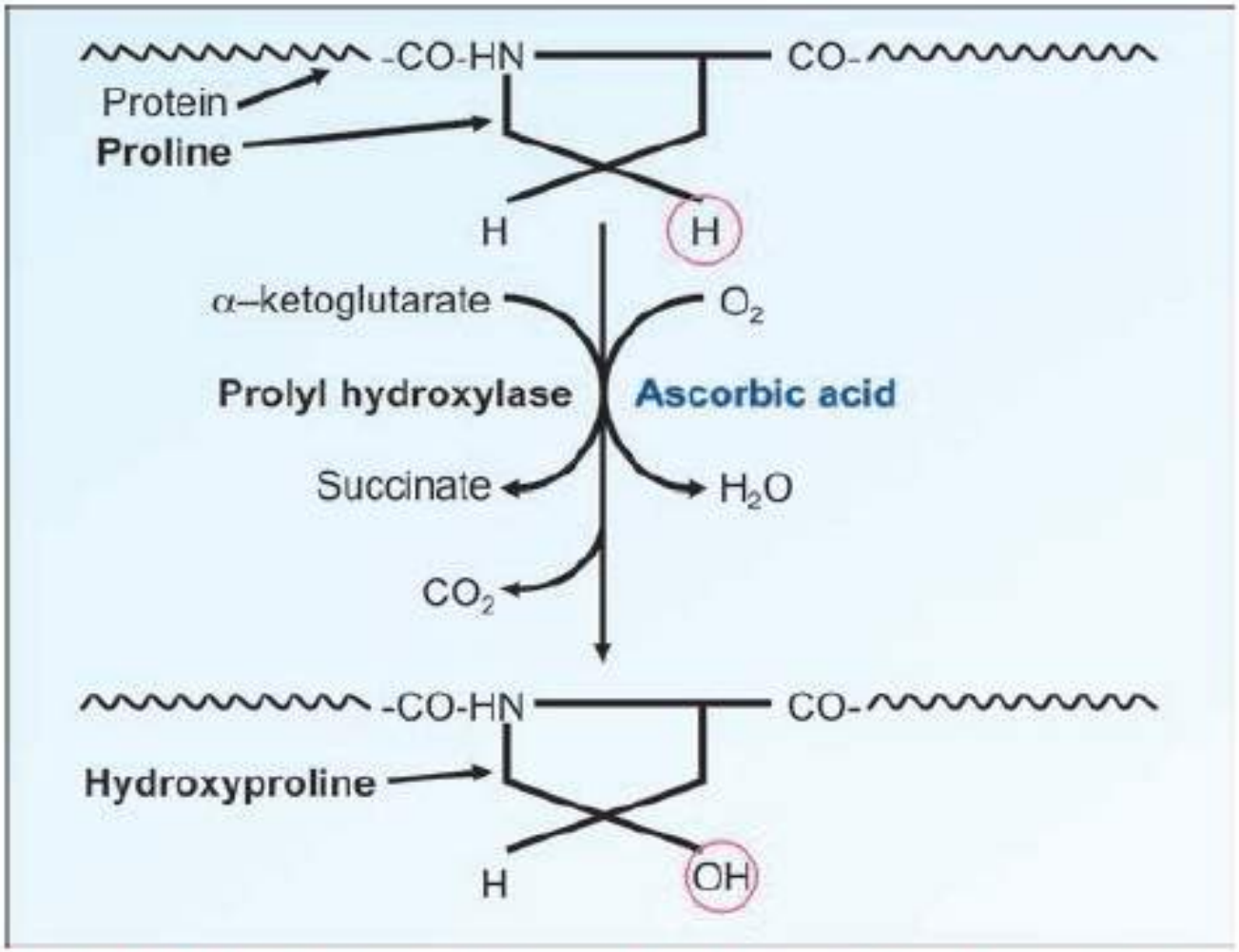


Fig. 34.29. Hydroxylation of proline to hydroxyproline needs ascorbic acid

Biochemical Functions of Vitamin C

1. Reversible oxidation-reduction

It can change between ascorbic acid and dehydro-ascorbic acid. Most of the physiological properties of the vitamin could be explained by this redox system.

2. Hydroxylation of proline and lysine

Ascorbic acid is necessary for the post-translational hydroxylation of proline and lysine residues (Fig. 34.29). Hydroxyproline and hydroxylysine are essential for the formation of cross links in the **collagen**, which gives the tensile strength to the fibres. This process is absolutely necessary for the normal production of supporting tissues such as osteoid, collagen and intercellular cement substance of capillaries.

3. Tryptophan metabolism

Ascorbic acid is necessary for the hydroxylation of tryptophan to 5-hydroxy tryptophan. This is required for the formation of serotonin (see Fig.17.10).

4. Tyrosine metabolism

Vitamin C helps in the oxidation of parahydroxy phenyl pyruvate to homogentisic acid (see Fig. 17.2).

5. Iron metabolism

Ascorbic acid enhances the iron absorption from the intestine (Chapter 35). Ascorbic acid reduces

ferric iron to ferrous state, which is preferentially absorbed.

6. Hemoglobin metabolism

It is useful for re-conversion of met-hemoglobin to hemoglobin.

7. Folic acid metabolism

Ascorbic acid is helping the enzyme folate reductase to reduce folic acid to tetrahydrofolic acid (Fig. 34.17). Thus it helps in the maturation of RBC.

8. Steroid synthesis

Large quantities of vitamin C are present in adrenal cortex. The ascorbic acid is depleted by ACTH stimulation. So the vitamin has some role in adrenal steroidogenesis. Vitamin C helps in the synthesis of bile acids from cholesterol. The initial 7-alpha-hydroxylase step is stimulated by the vitamin.

9. Phagocytosis

Ascorbic acid stimulates phagocytic action of leukocytes and helps in the formation of antibodies.

10. Anti-oxidant property

As an anti-oxidant (Chapter 20), it may prevent cancer formation. Aniline dyes are known to induce bladder cancer in factory workers. Daily intake of vitamin C reduces this risk for cancer.

11. Cataract

Vitamin C is concentrated in the lens of eye. Regular intake of ascorbic acid reduces the risk of cataract formation.

Deficiency Manifestations of Vitamin C

1. Scurvy

Gross deficiency of vitamin C results in scurvy.

2. Infantile scurvy (Barlow's disease)

In infants between 6 to 12 months of age, (period in which weaning from breast milk), the diet should be supplemented with vitamin C sources. Otherwise, deficiency of vitamin C is seen.

3. Hemorrhagic tendency

In ascorbic acid deficiency, **collagen is abnormal** and the intercellular cement substance is brittle (Fig. 34.29). So capillaries are fragile, leading to the tendency to bleed even under minor pressure.

Subcutaneous hemorrhage may be manifested as **petechiae** in mild deficiency and as **ecchymoses** or even hematoma in severe conditions. If a sphygmomanometer cuff is placed in the forearm, and the pressure is kept for 5 minutes, petechiae may be seen under the skin. This is a useful clinical test.

4. Internal hemorrhage

In severe cases, hemorrhage may occur in the conjunctiva and retina. Internal bleeding may be seen as epistaxis, hematuria or melena.

5. Oral cavity

In severe cases of scurvy, the gum becomes painful, swollen and spongy (Fig. 34.27A). The pulp is separated from the dentine and finally teeth are lost. Wound healing may be delayed.

6. Bones

In the bones, the deficiency results in the failure of the osteoblasts to form the intercellular substance, **osteoid**. Without the normal ground substance, the deposition of bone is arrested. The resulting scorbutic **bone is weak** and fractures easily. There may be hemorrhage into joint cavities. Painful swelling of joints may prevent locomotion of the patient.

Vitamin C and vitamin B are essential nutrients to maintain bone density and bone quality. Recent literature clearly shows that vitamin C and B affect bone quality determinant "collagen cross-link formation". Mildly elevated plasma homocysteine levels induced by vitamin B insufficiency and methylenetetrahydrofolate reductase (MTHFR) deteriorate normal collagen cross-link formation.

7. Anemia

In vitamin C deficiency, microcytic, hypochromic **anemia** is seen. Poikilocytosis and anisocytosis are also common in anemia due to deficiency of vitamin C. The reasons for anemia may be:

a. Loss of blood by hemorrhage
b. Decreased iron absorption
c. Decreased tetrahydrofolic acid
d. Accumulation of met-hemoglobin.

Dietary Sources of Vitamin C

Rich sources are amla (Indian gooseberry) (700 mg/100 g), guava (300 mg/100 g), lime, lemon and green leafy vegetables (Figs 34.27B and C).

Requirement of Vitamin C

Recommended daily allowance is 75 mg/day (equal to 50 ml orange juice). During pregnancy, lactation, and in aged people requirement may be 100 mg/day.

Therapeutic Use of Vitamin C

i. Vitamin C is used as an adjuvant in infections. Beneficial effect of ascorbic acid is reported in the treatment of tuberculosis, when plasma level is kept near to saturation point. Clinical dose is 500 mg per day.

ii. Because of its power to **heal wounds**, vitamin C has been recommended for treatment of ulcer, trauma and burns. Except in scurvy and sub-scorbutic conditions the therapeutic use of vitamin is not specific.

Megadose of Vitamin C

Linus Pauling (Nobel laureate) advocated megadoses (2000-5000 mg) of vitamin to prevent infections. But there is no clear evidence to support this theory. Classical picture of scurvy did not include common cold or respiratory infections as major features. There is experimental evidence in animals that ageing process is prevented by megadoses of vitamin C.

Toxicity of Vitamin C

Since it is a water soluble substance, excess vitamin C is excreted, and not accumulated in the body. However, more than 2000 mg of vitamin C daily for a long time can cause iron overload, because vitamin C helps in absorption of iron.